

FortiAnalyzer Key Concepts

This section provides information about basic FortiAnalyzer concepts and terms. If you are new to FortiAnalyzer, use this section to quickly understand this document and the FortiAnalyzer platform.

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- [FortiView dashboard on page 49](#)

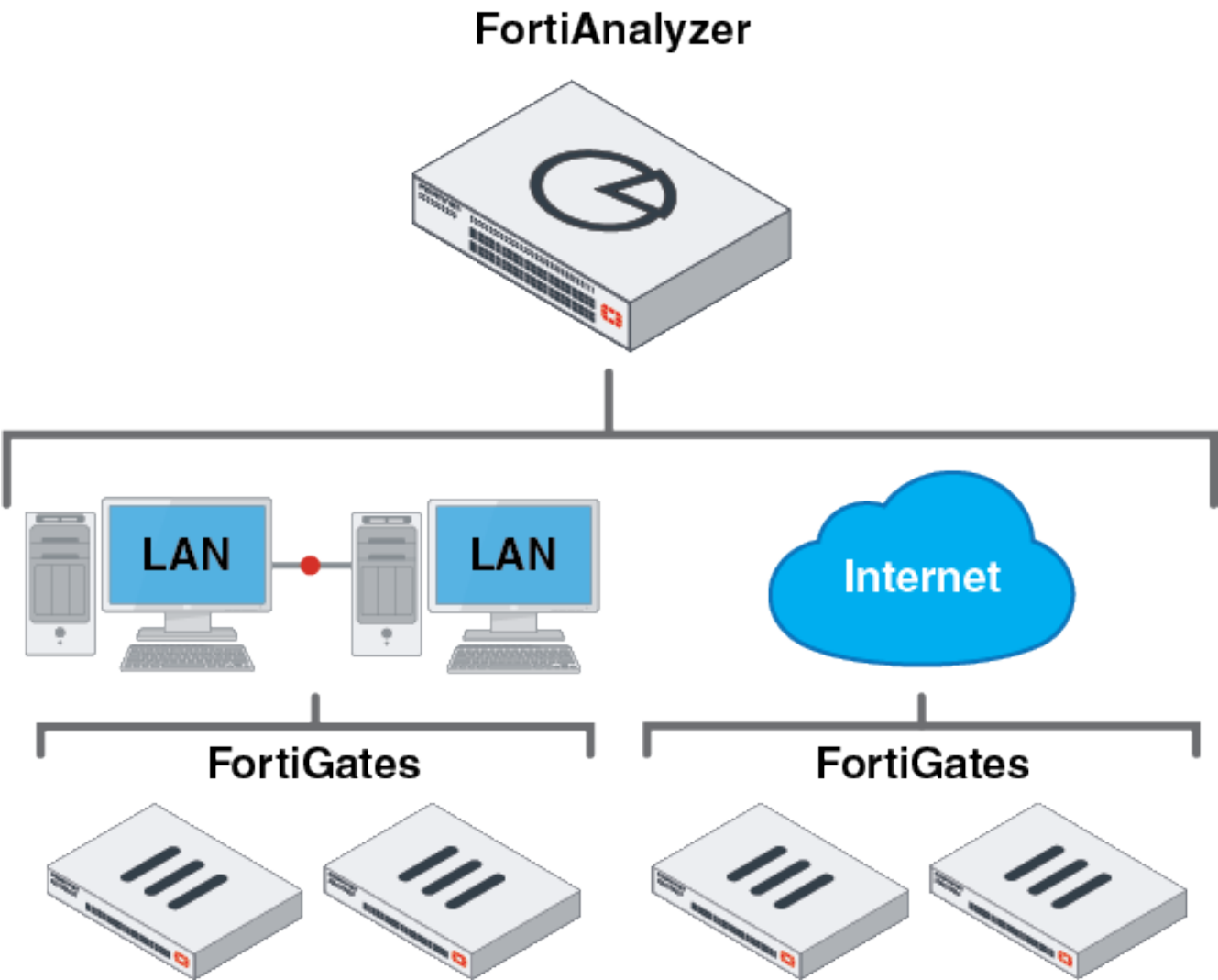
Operation modes

FortiAnalyzer can run in two operation modes: Analyzer and Collector. Choose the operation mode for your FortiAnalyzer units based on your network topology and requirements.

Analyzer mode

Analyzer mode is the default mode that supports all FortiAnalyzer features. Use this mode to aggregate logs from one or more Collectors.

The following diagram shows an example of deploying FortiAnalyzer in Analyzer mode.



Collector mode

When FortiAnalyzer is in Collector mode, its primary task is forwarding logs of the connected devices to an Analyzer and archiving the logs. Instead of writing logs to the database, the Collector retains logs in their original binary format for uploading. In this mode, most features are disabled.

Analyzer and Collector feature comparison

Feature	Analyzer Mode	Collector Mode
Device Manager	Yes	Yes

Feature	Analyzer Mode	Collector Mode
FortiView	Yes	No
Log View	Yes	Raw archive logs only
Incidents & Alerts	Yes	No
Monitoring devices	Yes	No
Reporting	Yes	No
System Settings	Yes	Yes
Log Forwarding	Yes	Yes

Analyzer–Collector collaboration

You can deploy Analyzer mode and Collector mode on different FortiAnalyzer units and make the units work together to improve the overall performance of log receiving, analysis, and reporting. The Analyzer offloads the log receiving task to the Collector so that the Analyzer can focus on data analysis and report generation. This maximizes the Collector's log receiving performance.

For an example of setting up Analyzer–Collector collaboration, see [Collectors and Analyzers on page 656](#).

FortiAnalyzer Fabric

FortiAnalyzer can also join a FortiAnalyzer Fabric which enables centralized viewing of devices, incidents, and alerts across multiple FortiAnalyzers acting as members.

The FortiAnalyzer Fabric is ideal for use in high volume environments with many FortiAnalyzers.

In this mode, FortiAnalyzer Fabric members form a Fabric with one device operating in supervisor mode as the root device. Incident, event, and log information is synced from members to the supervisor using the API.

See the [FortiAnalyzer Fabric Deployment Guide](#) for more information.

Administrative domains

Administrative domains (ADOMs) enable the admin administrator to constrain the access privileges of other FortiAnalyzer unit administrators to a subset of devices in the device list. For Fortinet devices with virtual domains (VDOMs), ADOMs can further restrict access to only data from a specific VDOM for a device.

Enabling ADOMs alters the available functions in the GUI and CLI. Access to the functions depends on whether you are logged in as the admin administrator. If you are logged in as the admin administrator, you can access all ADOMs. If you are not logged in as the admin administrator, the settings in your administrator account determines access to ADOMs.

For information on enabling and disabling ADOMs, see [Enabling and disabling the ADOM feature on page 537](#). For information on working with ADOMs, see [Administrative Domains \(ADOMs\) on page 533](#). For information on configuring administrator accounts, see [Managing administrator accounts on page 592](#).

Logs

Logs in FortiAnalyzer are in one of the following phases.

- Real-time log: Log entries that have just arrived and have not been added to the SQL database. These logs are stored in Archive in an uncompressed file.
- Archive logs: When a real-time log file in Archive has been completely inserted, that file is compressed and considered to be offline.
- Analytics logs or historical logs: Indexed in the SQL database and online.

In order for FortiAnalyzer to accept logs, the sending device must be registered in FortiAnalyzer. You can add devices to FortiAnalyzer by specifying the serial number and other details, or you may point the device's log settings to the FortiAnalyzer. If initiated by the remote device, the device must be authorized before logs can be received on FortiAnalyzer. See [Adding devices on page 93](#).

For more information on the types of logs collected for each device, see [Types of logs collected for each device on page 152](#).

Log encryption

Beginning in FortiAnalyzer 6.2, all logs from Fortinet devices (using Fortinet's proprietary protocol: OFTP) must be encrypted. FortiAnalyzer encryption level must be equal or less than the sending device's level. For example, when configuring logging from a FortiGate, FortiAnalyzer must have the same encryption level or lower than FortiGate in order to accept logs from FortiGate.

To configure the encryption level on FortiAnalyzer:

1. In the FortiAnalyzer CLI, enter the following commands:

```
config system global
    set enc-algorithm {high | low | medium}
```

To configure the encryption level on FortiGate:

1. In the FortiGate CLI, enter the following commands:

```
config log fortianalyzer setting
    set enc-algorithm {high-medium | high | low}
```

See also [Appendix B - Log Integrity and Secure Log Transfer on page 661](#).

Log storage

Logs and files are stored on the FortiAnalyzer disks. Logs are also temporarily stored in the SQL database.

You can configure data policy and disk utilization settings for devices. These are collectively called log storage settings.

You can configure global log and file storage settings. These apply to all logs and files in the FortiAnalyzer system regardless of log storage settings.

Log rolling

When FortiAnalyzer receives a log, it is stored in a file. Logs will continue to populate this file until its limit is reached, at which time the file is "rolled" which involves compressing the file and creating a new one for further logs of that type. There are two settings that you can use to configure when log rolling occurs, and both may be used at the same time, with rolling taking place when either condition is met.

- Log file size: This is enabled by default and set to 200 MB.
- At a scheduled time: Either daily or weekly at a set time.

Rolling the files daily is recommended to avoid a file from spanning more than 24 hours and masking the actual amount of days you are storing logs for.

See also [Configuring rolling and uploading of logs using the GUI on page 578](#).

Log deletion

When you reach your archive retention limit as defined by allocated storage size or specified days, FortiAnalyzer deletes old logs to make room for new logs. FortiAnalyzer can only delete files, not logs within a file. Controlling file growth is important because storage capacity is not infinite and it directly affects how old logs are deleted to make room for new logs.

FortiAnalyzer will delete old files based on which condition is forcing the deletion:

- Days: Delete the log file that contains logs which are *all* outside the configured day retention period. Log files can span several days, or even months. When this is the case, the file will not be considered eligible for deletion when logs that are within the configured retention days would be deleted. This can lead to Archive indicating it is storing more days than it is configured for (for example, 100/90 days). This is due to the number displaying the oldest log date, and not specifically that it has logs for each day up to that number.
- Storage size: Delete the log file with the oldest *last received* log. This can lead to the administrator not seeing the true amount of logs in analytics since there's no way to indicate that there are no logs for days 60 through 89, only that there are some logs from 90 days ago.

See also [Data policy and automatic deletion on page 48](#) and [Disk utilization for Archive and Analytic logs on page 48](#).

SQL database

FortiAnalyzer supports Structured Query Language (SQL) for logging and reporting. The log data is inserted into the SQL database to support data analysis in *FortiView*, *Log View*, and *Reports*. Remote SQL databases are not supported.

For more information, see [FortiView on page 107](#), [Types of logs collected for each device on page 152](#), and [Reports on page 459](#).

The log storage settings define how much FortiAnalyzer disk space to use for the SQL database.



When FortiAnalyzer is in Collector mode, the SQL database is disabled by default. If you want to use logs that require SQL when FortiAnalyzer is in Collector mode, you must enable the SQL database. See [Operation modes on page 42](#).

Analytics and Archive logs

Logs in FortiAnalyzer are in one of the following phases.

- Real-time log: Log entries that have just arrived and have not been added to the SQL database. These logs are stored in Archive in an uncompressed file.
- Archive logs: When a real-time log file in Archive has been completely inserted, that file is compressed and considered to be offline.
- Analytics logs or historical logs: Indexed in the SQL database and online.

Use a data policy to control how long to retain Analytics and Archive logs.

- [Archive logs on page 47](#)
- [Analytic logs on page 48](#)

Archive logs

When FortiAnalyzer receives a log, it is stored in a file. Logs will continue to populate this file until its limit is reached, at which time the file is "rolled" which involves compressing the file and creating a new one for further logs of that type. These files (rolled or otherwise) count against the archive retention limits and are referred to as *Archived* or *Offline* logs.

You cannot immediately view details about these logs in the *FortiView*, *Log View*, and *Incidents & Alerts* panes. You also cannot generate reports about the logs in the *Reports* pane.

Archive logs are stored unchanged and can be uploaded to a file server for use as backups.

- If you are using a FortiAnalyzer-VM, you may also choose to snapshot the data drive to backup your logs.
- If you are using a physical FortiAnalyzer which leverages RAID for storage, remember that RAID is not a backup solution.

Log storage in *Archive* is important since it is used to rebuild the database in the event of database corruption, or in some cases during upgrades.

Analytic logs

Immediately following the storage of a log in an archive, the same log is inserted into the SQL database. This function is also known as being *indexed*, and these logs are referred to as *Analytic* or *Online* logs.

Analytic logs are the only logs which are used for analysis in FortiAnalyzer *Log View* (excluding *Log Browse*), *Incidents and Events*, and *Reports*.

Analytic logs are dissected during insertion and any subtypes are stored as their own category. For example, security profile logs such as web filtering logs are sent and stored as Traffic logs when archived, however, Analytics extracts the relevant web filtering fields and stores them in a web filtering table.

Indexed logs take up significantly more space than the same amount of logs in Archive.

Most administrators may need to store between 30 and 60 days in Analytics, however, this should be configured for the amount of time that you would typically need to explore the logs for.

If you need to run analytics for dates outside your Analytics retention, you may perform a database rebuild and load the particular date range. A database rebuild involves purging all logs from Analytics and loading logs for the days of interest from Archive. Once analysis is complete, you can then rebuild once more to load the most current logs into analytics from the archive.

Data policy and automatic deletion

Use a data policy to control how long to keep compressed and indexed logs. When ADOMs are enabled, you can specify settings for each ADOM and the settings apply to all devices in that ADOM. When ADOMs are disabled, settings apply to all managed devices.

A data policy specifies:

- How long to keep Analytics logs indexed in the database
When the specified length of time in the data policy expires, logs are automatically purged from the database but remain compressed in a log file on the FortiAnalyzer disks.
- How long to keep Archive logs on the FortiAnalyzer disks
When the specified length of time in the data policy expires, Archive logs are deleted from the FortiAnalyzer disks.

See also [Log storage information on page 179](#).

Disk utilization for Archive and Analytic logs

You can specify how much of the total available FortiAnalyzer disk space to use for log storage. You can specify what ratio of the allotted storage space to use for logs that are indexed in the SQL database and for logs that are stored in a compressed format on the FortiAnalyzer disks. Then you can monitor how quickly device logs are filling up the allotted disk space.



Analytic logs indexed in the SQL database require more disk space than Archive logs (purged from the SQL database but remain compressed on the FortiAnalyzer disks).

An average Analytic log is 600 bytes, and an average Archive log is 80 bytes. By default, after seven days Analytic logs are compressed and are an average of 150 bytes.

Keep this difference in mind when specifying the storage ratio for Analytics and Archive logs.

When ADOMs are enabled, you can specify settings for each ADOM and the settings apply to all devices in that ADOM. When ADOMs are disabled, settings apply to all managed devices. See [Log storage information on page 179](#).

FortiView dashboard

FortiAnalyzer provides dashboards for Security Operations Center (SOC) administrators. FortiView includes monitors which enhance visualization for real-time activities and historical trends for analysts to effectively monitor network activities and security alerts. See [FortiView on page 106](#).

In high capacity environments, the FortiView module can be disabled to improve performance. See [Enabling and disabling FortiView on page 151](#).